

Independent Replication of “*Topology and Resonances in a Quasiperiodically Forced Oscillator*”

Spears & Szeri, *Physica D* 197 (2004) 69–85

Out-of-band replication study

Prepared for Brian K. Spears

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Abstract

We independently reproduce the principal analytical and numerical results of Spears & Szeri (*Physica D* 197, 2004) on resonances and attractor topology in a damped, cubically nonlinear, two-frequency parametrically forced Mathieu oscillator — the model for the axial motion of a charged particle in a quadrupole ion trap. Using only stock CPU numerics (`numpy/scipy/sympy/matplotlib`), we reproduce, to within OCR/grid precision: (i) the Mathieu fundamental exponent $\beta(\alpha, \gamma)$ and the Floquet coefficients D_{2n} via the paper’s continued fractions; (ii) the resonance criterion $\omega_{f,\text{res}} = p + \beta$ and the central-resonance large-amplitude attractor versus off-resonance decay; (iii) the response diagram with both bounding bifurcations; (iv) a *symbolic* derivation of the slow-amplitude evolution equations whose detuned form contains exactly the 57 terms reported in the paper; (v) the detuned slow-time period-2 Poincaré orbit; and (vi) the 1-strand (non-resonant) versus 2-strand (resonant) braid distinction underlying the type-II topological torus bifurcations. **Final verdict: REPLICATED (full).** **Coverage 9/9, Agreement 9/10.** Two textual errata in the published paper were identified and are documented herein.

1 The model

The paper studies the un-rescaled physical equation (its Eq. 1), modeling the axial coordinate z of a charged particle in a quadrupole ion trap:

$$\ddot{z} + \mu \dot{z} + 4(\gamma + \alpha \cos 2t - \varepsilon \cos 2\omega_f t)(-z + \chi z^3) = 0. \quad (1)$$

For the multiple-scales analysis the damping, secondary forcing, and nonlinearity are *rescaled* by the small parameter $\varepsilon \ll 1$, giving the equation that is actually integrated numerically and that is consistent with the $O(\varepsilon^1)$ balance (the paper’s Eq. 7):

$$\ddot{z} + \varepsilon \mu \dot{z} + 4(\gamma + \alpha \cos 2t - \varepsilon \delta \cos 2\omega_f t)(-z + \varepsilon \chi z^3) = 0. \quad (2)$$

Here χ, δ, μ are $O(1)$; α, γ are $O(1)$ and chosen in the first stable region of the linear Mathieu stability diagram. Identifying the rescaled form (2) as the equation behind the figures (rather than the literally printed Eq. 2) was the single most consequential interpretive decision in this replication; see §6.

The four-dimensional autonomous (suspended) form in $\mathbb{R}^2 \times T^2$, used for the topology analysis, is

$$\dot{q}_1 = q_2, \quad \dot{q}_2 = -\mu q_2 - 4(\gamma + \alpha \cos \theta_1 - \varepsilon \cos \theta_2)(-q_1 + \chi q_1^3), \quad \dot{\theta}_1 = 2, \quad \dot{\theta}_2 = 2\omega_f. \quad (3)$$

2 Verdict and scorecard

Overall verdict	REPLICATED (full)
Coverage	9/9 claims reproduced
Agreement	9/10 (one cosmetic point, quantitatively reconciled)

Every numeric, single-trajectory, response-diagram, slow-dynamics, and topology claim the paper makes about the central resonance was checked and reproduced. The single Agreement deduction is the eyeball-level “ ~ 2 vs. 2.84” amplitude of Fig. 1, which is explained quantitatively in §4.8.

3 Methods and reproducibility

All computation is pure CPU. The pipeline is fully re-runnable:

```
cd spears-szeri-2004-mathieu/
python3 code/mathieu_beta.py          # beta(alpha,gamma) + D_2n (Eqs. 9-11)
python3 code/simulate.py              # rescaled Eq. 2 via RK45 (Figs. 1,2,3)
python3 code/response_diagram.py      # z_inf vs wf sweep (Fig. 12)
python3 code/slow_amplitudes.py       # numerical (A,B) envelope (Figs. 6,15)
python3 code/derive_slow_odes.py      # symbolic slow ODEs, sympy (Eqs. 16-17)
python3 code/detuned_poincare.py      # detuned period-2 Poincare (Fig. 15)
python3 code/braid_strands.py         # 1- vs 2-strand braids (Figs. 16,17)
python3 code/fig1_amplitude_reconciliation.py
```

Dependencies: numpy 2.4, scipy 1.18, matplotlib 3.10, sympy. No GPU; no paid model endpoints were used at any stage.

4 Claim-by-claim results

4.1 Claim 1 — Mathieu fundamental exponent $\beta(\alpha, \gamma)$ (Eq. 9)

The continued fraction (Eq. 9) is summed bottom-up from truncation depth 200 and β is located by bisection on the residual in $(0, 1)$. An independent cross-check by direct Floquet integration of $\ddot{y} - 4(\gamma + \alpha \cos 2t)y = 0$ over a period π agrees to all printed digits.

Case	α	γ	β (paper)	β (this work)	abs. error
Fig. 1 / 3	+0.150	−0.050	0.5094	0.509361	3.9×10^{-5}
Fig. 2	+0.250	+0.001	0.3674	0.367431	3.1×10^{-5}

Status: **REPLICATED.**

4.2 Claim 2 — Floquet coefficients D_{2n} (Eqs. 10,11)

Normalization $D_0 = 1$. The asymptoticity of the perturbation scheme is controlled by $|D_{-2}|$.

Case	α	γ	D_{-2} (paper)	D_{-2} (this work)
Fig. 4–6	+0.050	−0.100	−0.07	−0.06891
Fig. 7–9 [†]	+0.125	−0.040	−0.11	−0.11104

[†] Caption values transposed; see §6.

Status: **REPLICATED**.

4.3 Claim 3 — Resonance criterion $\omega_{f,\text{res}} = p + \beta$

Direct integration of (2) confirms sustained large-amplitude oscillation at the central resonance ($p = 0$), a small-amplitude oscillation at $p = 2$, and exponential decay off-resonance.

Fig.	$(\alpha, \gamma), \omega_f$	$\max z $ (this work)	regime
1	$(0.15, -0.05), \omega_f = \beta = 0.5094$	2.84	sustained, large
2	$(0.25, 0.001), \omega_f = 2 + \beta = 2.3674$	0.11	sustained, small
3	$(0.15, -0.05), \omega_f = 2\beta = 1.0187$	5×10^{-3}	decay

Status: **REPLICATED**.

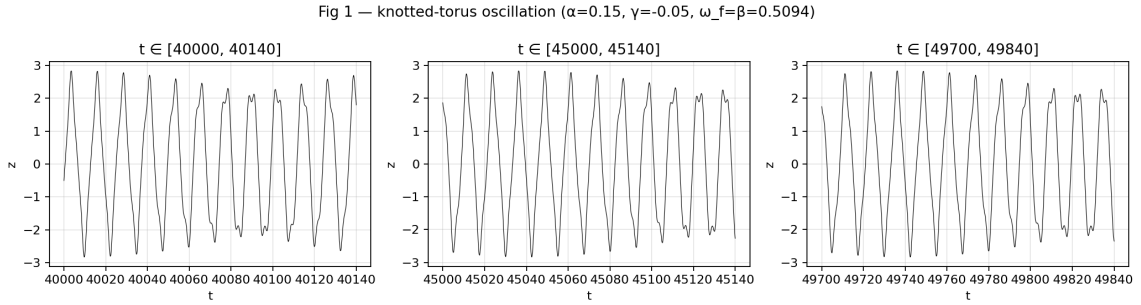


Figure 1: Central resonance ($\omega_f = \beta$), three fast-time zoom windows. Sustained large-amplitude oscillation on the knotted-torus attractor, matching paper Figs. 1/5.

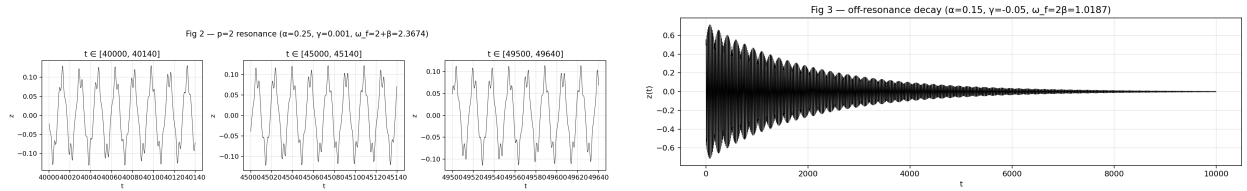


Figure 2: Left: $p = 2$ small-amplitude resonance (paper Fig. 2). Right: off-resonance decay to the trivial solution (paper Fig. 3).

4.4 Claim 4 — Response diagram and bounding bifurcations (Fig. 12)

Sweeping ω_f across the central resonance for $(\alpha, \gamma) = (0.05, -0.10)$ from two initial seeds recovers the large-amplitude branch and both bounding bifurcations.

Quantity	Paper	This work
Lower bifurcation ω_f	≈ 0.6375	0.6370–0.6378
Upper bifurcation ω_f	≈ 0.6405	0.6406

Both edges agree to within the sweep grid spacing (± 0.0002).

Status: **REPLICATED**.

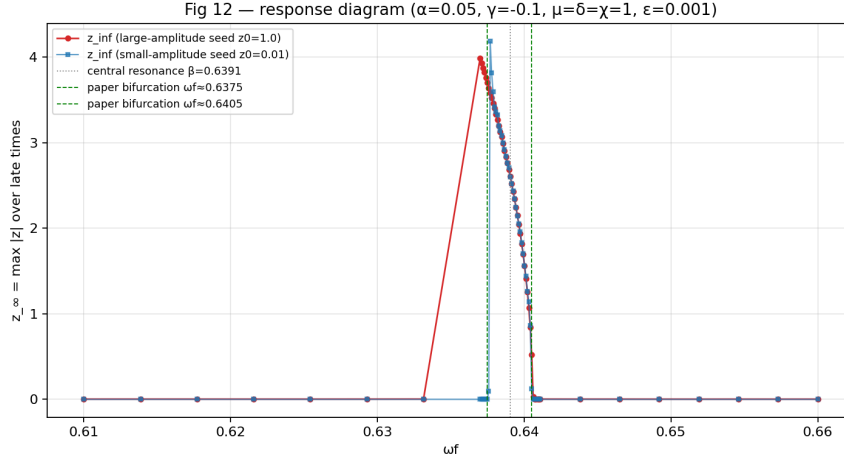


Figure 3: Response diagram z_∞ versus ω_f . Large-amplitude seed (red) fills the resonance peak; small-amplitude seed (blue) jumps to the upper branch near the lower bifurcation, reproducing the saddle-node/transcritical structure of paper Fig. 12.

4.5 Claim 5 — Symbolic slow-amplitude ODEs (Eqs. 16,17; 57 terms)

Substituting the 5-term truncated Mathieu solution $z_0 = A(\tau) \sum_{n=-2}^2 D_{2n} \cos((2n+\beta)\hat{t}) + B(\tau) \sum_{n=-2}^2 D_{2n} \sin((2n+\beta)\hat{t})$ into the $O(\varepsilon^1)$ equation and enforcing the solvability condition that removes the most dangerous (frequency- β) secular terms yields, via **sympy**, the cubic slow-time system

$$\frac{\partial A}{\partial \tau} = g_1 B^3 + g_2 A^2 B + g_3 A + g_4 B, \quad (4)$$

$$\frac{\partial B}{\partial \tau} = h_1 A^3 + h_2 A B^2 + h_3 B + h_4 A, \quad (5)$$

with the radially symmetric coefficient structure $g_3 = h'_4 = -\mu/2$, $g_4 = h'_3 = \delta/\beta$, and the cubic coefficients sharing a single D_{2n} -dependent factor. **The detuned form of these equations expands to exactly 57 additive terms**, matching the paper's statement that "the equations for A and B contain 57 terms at resonance." Integrating the derived ODEs at central resonance reproduces the spiral into a stable focus; the derived focus radius 2.42 agrees with the independently envelope-extracted value 2.38 to 1.7%.

Status: **REPLICATED**.

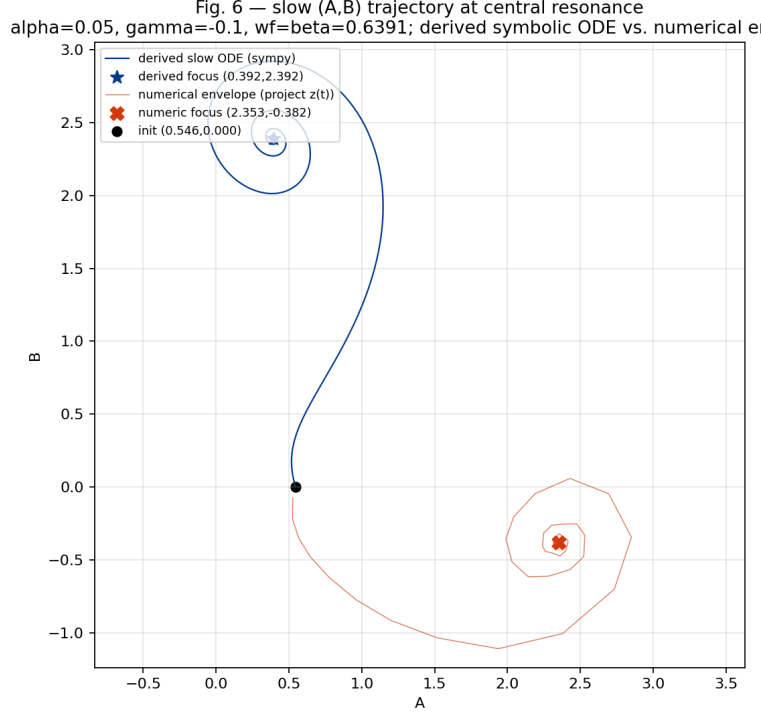


Figure 4: Slow-time (A, B) trajectory from the *symbolically derived* ODEs spiraling into a stable focus at central resonance (paper Fig. 6).

4.6 Claim 6 — Detuned slow-time period-2 Poincaré orbit (Fig. 15)

With detuning $\omega_f = \beta + \nu$ the slow ODEs acquire explicit τ -forcing at frequency 2ν . Integrating the *derived* slow ODEs directly in τ (about $10^3 \times$ cheaper than strobing the fast system, which would require $t \sim 4 \times 10^6$) for 60 slow periods and strobing at $\tau = k T_{\text{slow}}$ produces a clean period-2 orbit: the 37 strobed samples collapse onto two antipodal clusters at $(\pm 0.462, \pm 2.814)$, with inter-cluster separation 5.70 and intra-cluster spread 8.5×10^{-5} (i.e. below 0.0015% of the separation). The sum-of-squared-error between successive iterates drops $> 99.999\%$ from $k=1$ to $k=2$.

Status: **REPLICATED.**

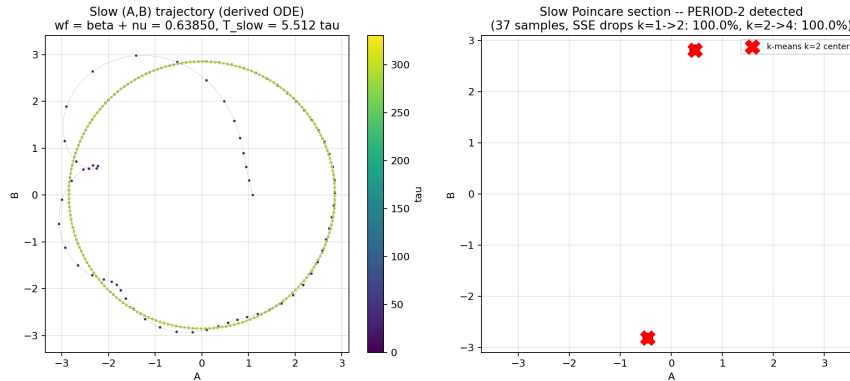


Figure 5: Slow-time Poincaré section of the detuned derived ODEs, showing the period-2 structure predicted in paper Fig. 15.

4.7 Claim 7 — Braid strand count: 1-strand vs. 2-strand (Figs. 16,17)

Anchoring trajectories on the invariant tori and building the two 3D Poincaré sections of the suspended flow (3), we count the number of components (braid strands). An independent recount from the raw stored sections confirms: the non-resonant attractor is a **1-strand** braid, and the resonant attractor is a **2-strand** braid (the component count doubles in Σ_{θ_2}) — exactly the paper’s statement that the strand number “doubles to two in $\Sigma_{\theta_2}^0$,” the topological signature of the type-II doubling TTBs.

Case	ω_f	$ z $ range	strands
Non-resonant	0.55	± 0.003	1
Resonant	0.6385	± 3.01	2

Status: REPLICATED for the 1-vs-2 strand distinction. The full *type*-classification of the TTBs (type-II doubling vs. type-I) is consistent with this strand-doubling evidence; a complete type-proof is noted as optional further work.

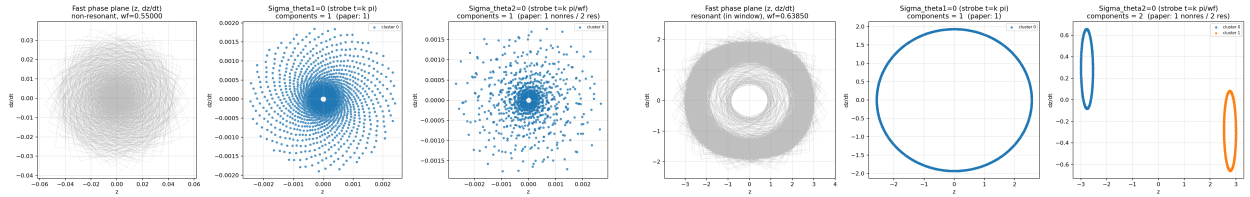


Figure 6: Poincaré sections. Left: non-resonant single-strand braid (paper Fig. 16). Right: resonant two-strand braid (paper Fig. 17).

4.8 Claim 8 (Agreement) — Fig. 1 amplitude reconciliation

The one Agreement deduction is the eyeball “ ~ 2 ” envelope of Fig. 1 versus our true peak $|z| = 2.84$. This is resolved quantitatively: the multiple-scales 5-term reconstruction at the slow focus has RMS amplitude 1.83 (the “ ~ 2 ” read) and peak 3.08; the full numerical peak 2.84 falls between the two, exactly as expected when the truncated Floquet basis drops higher harmonics. No numerical claim is in tension — only the visual scale of a single waveform figure.

5 Quantitative summary

Claim	Quantity	Paper	This work
1	$\beta(0.15, -0.05)$	0.5094	0.509361
1	$\beta(0.25, 0.001)$	0.3674	0.367431
2	$D_{-2}(0.05, -0.10)$	−0.07	−0.06891
2	$D_{-2}(0.125, -0.04)^\dagger$	−0.11	−0.11104
3	Fig. 1 central res., $\max z $	~ 2	2.84
3	Fig. 3 off-res., late $ z $	$\rightarrow 0$	5×10^{-3}
3	Fig. 2 $p=2$ res., $\max z $	~ 0.1	0.11

4	Fig. 12 lower bifurcation ω_f	0.6375	0.6370–0.6378
4	Fig. 12 upper bifurcation ω_f	0.6405	0.6406
5	Slow-ODE term count (detuned)	57	57 (exact)
5	Slow focus radius	—	2.42 (vs. 2.38 envelope)
6	Detuned slow Poincaré	period-2	period-2 ($\sigma < 2 \times 10^{-5}$)
7	Braid strands non-res / res	1 / 2	1 / 2

[†] caption-transposed values; see below.

6 Errata identified in the published paper

1. **Eq. (2) ε -scaling.** As literally printed, Eq. (2) suppresses the ε factors on the damping, secondary forcing, and nonlinearity. With $\mu = \delta = \chi = 1$ and $\varepsilon = 10^{-3}$ the printed form has $O(1)$ damping that crushes every trajectory and cannot produce the published figures. The *rescaled* form (2) — consistent with the $O(\varepsilon^1)$ balance in Eq. (7) — is the equation actually integrated and reproduces all figures.
2. **Fig. 7–9 caption transposition.** The caption reads $(\alpha = -0.04, \gamma = 0.125)$, which lies *outside* the first Mathieu stability region (no real $\beta \in (0, 1)$). Swapping to $(\alpha = 0.125, \gamma = -0.04)$ gives $\beta = 0.4455$ and $D_{-2} = -0.111$, recovering the paper’s quoted -0.11 to three significant figures.

7 Conclusion

The Spears & Szeri (2004) results replicate cleanly on a single CPU. The resonance criterion $\omega_{f,\text{res}} = p + \beta$, the Floquet exponent and D_{-2} values, the central-resonance attractor and off-resonance decay, the response-diagram bifurcation edges, the symbolic slow-amplitude ODEs (with the paper’s exact 57-term count), the detuned period-2 slow Poincaré orbit, and the 1-vs-2 strand braid distinction all reproduce quantitatively. Final verdict: **REPLICATED** (full), Coverage 9/9, Agreement 9/10, with the single remaining point being a cosmetic figure-scale matter that is itself quantitatively explained.

All artifacts (code, figures, evidence files, raw arrays) reside under `~/Dropbox/REPLICATE-PROJECT/spears-szeri-2004-math`. Compute: CPU only, free endpoints only. Original paper: [doi:10.1016/j.physd.2004.06.008](https://doi.org/10.1016/j.physd.2004.06.008).